PROJECT AND TEAM INFORMATION

Project Title

(Try to choose a catchy title. Max 20 words).

Smart Warehouse Resource Management and Robot Task Scheduling System

Student / Team Information

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PROPOSAL DESCRIPTION (10 pts)

Motivation (1 pt)

Modern warehouses handle a large number of customer orders, products, inventory records, and warehouse operations. When these activities are managed manually or independently, problems such as inefficient task scheduling, resource conflicts, delayed order processing, incorrect inventory updates, and poor utilization of warehouse resources can occur.

The proposed ***Smart Warehouse Resource Management and Robot Task Scheduling System*** *aims to provide a software-based solution for managing these activities in an organized and efficient manner. The system will simulate warehouse operations in which customer orders are converted into warehouse tasks such as product picking, packing, and dispatching. These tasks will be managed using Operating System concepts such as process states, ready queues, CPU scheduling, resource allocation, synchronization, mutual exclusion, and deadlock detection and recovery.*

A Database Management System will maintain information about customers, orders, products, inventory, robots, tasks, and execution records. In addition, an AI module will analyse historical order data to predict product demand and recommend task priorities.

The project is important because it demonstrates how DBMS, OS, and AI can work together to solve a practical warehouse-management problem rather than implementing these technologies as separate modules.

State of the Art / Current solution (1 pt)

In modern warehouses, inventory and order information is commonly managed using Warehouse Management Systems (WMS) and database-based applications. These systems maintain product information, inventory levels, orders, storage locations, and shipment records. Larger warehouses may also use automated machines and robots for activities such as transportation, picking, and sorting.

However, these systems generally focus on operational warehouse management and automation rather than demonstrating the interaction between Operating System resource-management concepts and a warehouse environment. Task scheduling, resource allocation, synchronization, and deadlock handling are usually hidden inside complex software and are not presented as an educational simulation.

The proposed project provides a simplified and transparent model of warehouse operations. It connects database management with OS concepts such as scheduling, processes, resource allocation, synchronization, and deadlock handling. An AI component is also incorporated to analyse historical order data, predict demand, and recommend task priorities.

Project Goals and Milestones (2 pts)

*The main goal of this project is to develop a software-based Smart Warehouse Resource Management and Robot Task Scheduling System that integrates **DBMS, Operating System concepts, and AI** to simulate efficient warehouse operations.*

The project will achieve the following goals:

1. *Develop a database to manage customers, products, inventory, orders, warehouse tasks, robots, resources, and execution logs.*
2. *Convert customer orders into warehouse tasks such as picking, packing, and dispatching.*
3. *Implement an OS-based task scheduling module using algorithms such as **FCFS, SJF, Priority Scheduling, and Round Robin** and compare their scheduling performance.*
4. *Implement resource management for warehouse robots and simulate robot allocation to tasks.*
5. *Implement synchronization and mutual exclusion to prevent multiple tasks from incorrectly accessing the same robot or shared resource.*
6. *Implement deadlock detection and a basic recovery mechanism for tasks competing for multiple resources.*
7. *Integrate an AI module that uses historical order data to predict product demand and provide task-priority recommendations.*
8. *Maintain consistency between task execution and the warehouse database using transactions and appropriate database operations.*
9. *Provide an administrator dashboard showing inventory, orders, robot status, task status, scheduling results, and system reports.*

Milestones

Milestone 1: *Requirement analysis and system design.*

Milestone 2: *Database design and implementation.*

Milestone 3: *Order, inventory, and warehouse-task modules.*

Milestone 4: *OS scheduling and resource-management modules.*

Milestone 5: *Synchronization and deadlock detection/recovery.*

Milestone 6: *AI demand-prediction and priority-recommendation module.*

Milestone 7: *Integration, testing, performance comparison, and final dashboard.*

Project Approach (3 pts)

The proposed system will be developed as a modular software application consisting of DBMS, Operating System, and AI components. The system will follow a layered approach so that each technology has a clearly defined responsibility while working together as one warehouse-management system.

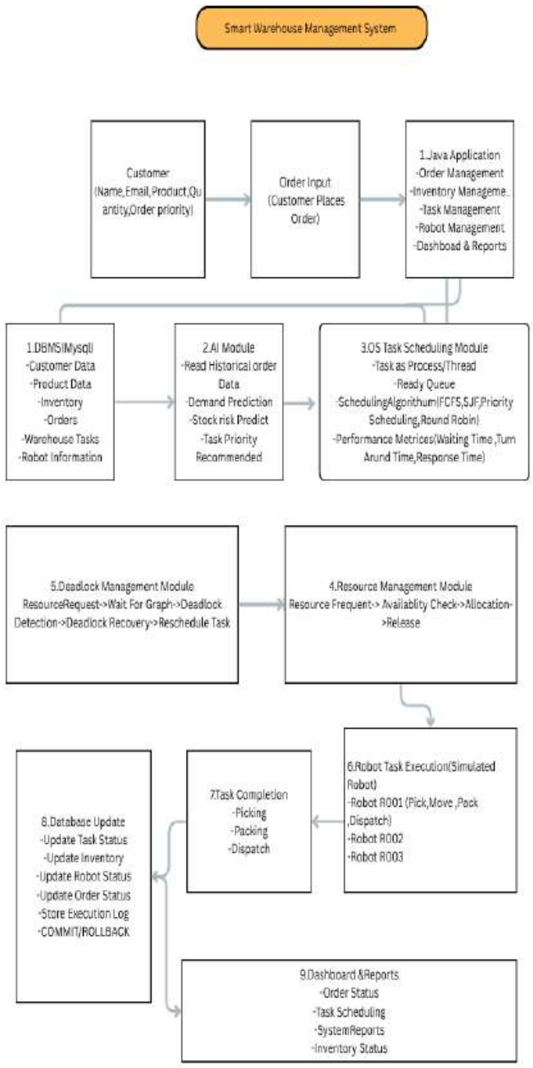
The ***DBMS layer*** *will use MySQL to store and manage customer information, products, inventory, orders, warehouse tasks, robot resources, scheduling records, and execution logs. SQL operations such as SELECT, INSERT, and UPDATE will be used for data management. Database transactions, COMMIT, ROLLBACK, and consistency mechanisms will be used where multiple related operations must be completed safely.*

The ***OS simulation layer*** *will represent warehouse tasks as processes/threads. Tasks generated from customer orders will enter a ready queue and will be scheduled using FCFS, SJF, Priority Scheduling, and Round Robin. The system will calculate scheduling metrics such as waiting time, turnaround time, and response time. Resource allocation will simulate robots as shared resources. Synchronization, mutual exclusion, critical sections, and locks will prevent resource conflicts. A wait-for graph can be used for deadlock detection and a recovery mechanism will allow affected tasks to be rescheduled.*

The ***AI layer*** *will use historical order data retrieved from the database to predict product demand and identify stock-risk patterns. The prediction will be used to recommend task priorities, while the OS scheduling module will perform the actual task scheduling.*

The main application will be developed in ***Java****, MySQL will be used as the database, and* ***Python with machine-learning libraries*** *will be used for the AI component. JDBC will provide communication between the Java application and MySQL. JavaScript will not be used.*

System Architecture (High Level Diagram)(2 pts)



Project Outcome / Deliverables (1 pts)

The project will deliver a working software prototype of a Smart Warehouse Resource Management and Robot Task Scheduling System integrating DBMS, Operating System concepts, and AI.

The major deliverables will include:

1. *A MySQL database for customers, products, inventory, orders, warehouse tasks, robots, resources, and execution logs.*
2. *A Java-based application for customer order entry, warehouse task management, resource allocation, and system monitoring.*
3. *An OS-based task scheduling module implementing FCFS, SJF, Priority Scheduling, and Round Robin with scheduling performance metrics.*
4. *A resource-management module that simulates robot allocation, synchronization, mutual exclusion, and deadlock detection/recovery.*
5. *An AI module using historical order data to predict product demand, identify stock-risk patterns, and recommend task priorities.*
6. *Integration between the AI recommendations, OS scheduling module, and database.*
7. *An administrator dashboard showing order status, inventory levels, robot availability, task execution, scheduling results, and system reports.*
8. *Final documentation containing system design, database schema, algorithms, implementation details, testing results, and project outcomes.*

Assumptions

The warehouse will be represented as a software simulation rather than a physical warehouse. Robots will be treated as software-simulated shared resources. Product inventory and order information will be maintained in MySQL. Historical order data required for AI prediction will be available in the database or generated as sample training data. The AI module will provide recommendations rather than directly controlling physical robots. Network communication and physical robot hardware are outside the project scope.

References

1. *Abraham Silberschatz, Peter B. Galvin, Greg Gagne,* ***Operating System Concepts****, Wiley.*
2. *Ramez Elmasri, Shamkant B. Navathe,* ***Fundamentals of Database Systems****, Pearson.*
3. *Ian Sommerville,* ***Software Engineering****, Pearson.*
4. *Stuart Russell and Peter Norvig,* ***Artificial Intelligence: A Modern Approach****, Pearson.*
5. *MySQL Documentation — for database implementation and SQL concepts.*
6. *Oracle Java Documentation — for Java programming and concurrency concepts.*
7. *Scikit-learn Documentation — for machine-learning implementation.*